

Model Performance Report: Predicting Irrigation Need

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1 Introduction

The objective of this project was to develop a predictive model to determine the **Irrigation Need** based on agricultural and environmental features. This report evaluates several machine learning approaches, ranging from unsupervised clustering to supervised classification, as documented in the competition submission history.

2 Model Performance Summary

The models were evaluated based on their Public Accuracy Score. The following table summarizes the performance of the various algorithms tested, as referenced in the file `submissions.png`.

Model Algorithm	Status	Public Accuracy Score
Decision Tree	Completed	0.97500
Submission (3)	Completed	0.94222
Submission (2) / Hello (1)	Completed	0.93573
Gaussian Naive Bayes	Completed	0.76479
K-Means Clustering	Completed	0.34826
Linear Regression	Completed	0.15905

Table 1: Comparison of model accuracy scores across different methodologies.

3 Methodology and Analysis

3.1 Top Performer: Decision Tree

The **Decision Tree** model achieved the highest accuracy of **97.5%**. This suggests that the relationship between features and irrigation requirements is non-linear and hierarchical. This model effectively captured the specific logic needed for classification, outperforming all other attempts.

3.2 Gaussian Naive Bayes

The Gaussian Naive Bayes approach yielded a score of **0.7648**. While significantly better than random guessing, its performance was likely hindered by the "independence assumption," as environmental features in irrigation data are often highly correlated.

3.3 Unsupervised and Linear Approaches

- **K-Means Clustering:** Achieved a score of **0.3483**. This indicated that while data patterns exist, clustering alone is insufficient for precise classification without target labels.

- **Linear Regression:** Produced the lowest score (**0.1591**), confirming that irrigation need is a discrete classification task rather than a continuous linear trend.

4 Conclusion

The experimental results demonstrate that tree-based supervised learning is the most effective approach for this dataset. The high accuracy of the Decision Tree (97.5%) provides a robust baseline for real-world irrigation prediction deployment.